

UI24CS37

Assignment 1

OS

1. Shell script Program to accept a character and check whether it is an

(i) Lower case alphabet

(ii) Upper case alphabet

(iii) A digit

(iv) Special symbol

(v) Vowel

Code :

```
read -p "Enter a single character: " ch

if [[ $ch =~ [a-z] ]]; then
    echo "Lowercase alphabet"
    case $ch in
        [aeiou]) echo "It is a vowel." ;;
        *) echo "It is not a vowel." ;;
    esac
elif [[ $ch =~ [A-Z] ]]; then
    echo "Uppercase alphabet"
    case $ch in
        [AEIOU]) echo "It is a vowel." ;;
        *) echo "It is not a vowel." ;;
    esac
elif [[ $ch =~ [0-9] ]]; then
    echo "Digit"
else
    echo "Special Symbol"
fi
```

```

keyur@keyur-VirtualBox:~/os/assignment1$ bash 1.sh
Enter a single character: 5
Digit
keyur@keyur-VirtualBox:~/os/assignment1$ bash 1.sh
Enter a single character: g
Lowercase alphabet
It is not a vowel.
keyur@keyur-VirtualBox:~/os/assignment1$ bash 1.sh
Enter a single character: O
Uppercase alphabet
It is a vowel.
keyur@keyur-VirtualBox:~/os/assignment1$ bash 1.sh
Enter a single character: *
Special Symbol

```

2. Using case control structure Using case .. esac structure

(i) Find the number of users logged into the system

(ii) Print the calendar for the current year

(iii) Print the date

Code :

```

echo "Choose an option:\n"
echo "1. Number of users logged in"
echo "2. Calendar for the current year"
echo "3. Print today's date"
read -p "Enter your choice (1/2/3): " choice

case $choice in
    1) who | wc -l ;;
    2) cal -y ;;
    3) date ;;
    *) echo "Invalid choice" ;;
esac

```

```
keyur@keyur-VirtualBox:~/os/assignment1$ gedit 2.sh
keyur@keyur-VirtualBox:~/os/assignment1$ bash 2.sh
Choose an option:\n
1. Number of users logged in
2. Calendar for the current year
3. Print today's date
Enter your choice (1/2/3): 1
2
```

```
keyur@keyur-VirtualBox:~/os/assignment1$ bash 2.sh
Choose an option:\n
1. Number of users logged in
2. Calendar for the current year
3. Print today's date
Enter your choice (1/2/3): 2

                2025
      January      February      March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
      1  2  3  4          1          1
  5  6  7  8  9 10 11    2  3  4  5  6  7  8    2  3  4  5  6  7  8
12 13 14 15 16 17 18    9 10 11 12 13 14 15    9 10 11 12 13 14 15
19 20 21 22 23 24 25   16 17 18 19 20 21 22   16 17 18 19 20 21 22
26 27 28 29 30 31     23 24 25 26 27 28     23 24 25 26 27 28 29
                                30 31

      April      May      June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
      1  2  3  4  5          1  2  3    1  2  3  4  5  6  7
  6  7  8  9 10 11 12    4  5  6  7  8  9 10    8  9 10 11 12 13 14
13 14 15 16 17 18 19   11 12 13 14 15 16 17   15 16 17 18 19 20 21
20 21 22 23 24 25 26   18 19 20 21 22 23 24   22 23 24 25 26 27 28
27 28 29 30          25 26 27 28 29 30 31   29 30

      July      August      September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
      1  2  3  4  5          1  2          1  2  3  4  5  6
  6  7  8  9 10 11 12    3  4  5  6  7  8  9    7  8  9 10 11 12 13
13 14 15 16 17 18 19   10 11 12 13 14 15 16   14 15 16 17 18 19 20
20 21 22 23 24 25 26   17 18 19 20 21 22 23   21 22 23 24 25 26 27
27 28 29 30 31     24 25 26 27 28 29 30   28 29 30
21
```

```

      1  2  3  4          1      1  2  3  4  5  6
5  6  7  8  9 10 11    2  3  4  5  6  7  8    7  8  9 10 11 12 13
12 13 14 15 16 17 18   9 10 11 12 13 14 15   14 15 16 17 18 19 20
19 20 21 22 23 24 25  16 17 18 19 20 21 22   21 22 23 24 25 26 27
26 27 28 29 30 31     23 24 25 26 27 28 29   28 29 30 31
                        30
keyur@keyur-VirtualBox:~/os/assignment1$ bash 2.sh
Choose an option:\n
1. Number of users logged in
2. Calendar for the current year
3. Print today's date
Enter your choice (1/2/3): 3
Sun Aug 10 03:48:02 PM IST 2025
keyur@keyur-VirtualBox:~/os/assignment1$

```

3. Shell Script Program to check whether the given file is a directory or not.

Code :

```

read -p "Enter file/directory name: " file

if [ -d "$file" ]; then
    echo "$file is a directory"
else
    echo "$file is not a directory"
fi

```

```

keyur@keyur-VirtualBox:~/os/assignment1$ gedit 3.sh
keyur@keyur-VirtualBox:~/os/assignment1$ bash 3.sh
Enter file/directory name: a1
a1 is a directory
keyur@keyur-VirtualBox:~/os/assignment1$ bash 3.sh
Enter file/directory name: 1.sh
1.sh is not a directory

```

4. Shell Script Program to count the number of files in a Directory.

Code :

```

read -p "Enter directory path: " dir

if [ -d "$dir" ]; then
    count=$(find "$dir" -maxdepth 1 -type f | wc -l)
    echo "Number of files: $count"
else
    echo "Directory does not exist"
fi

```

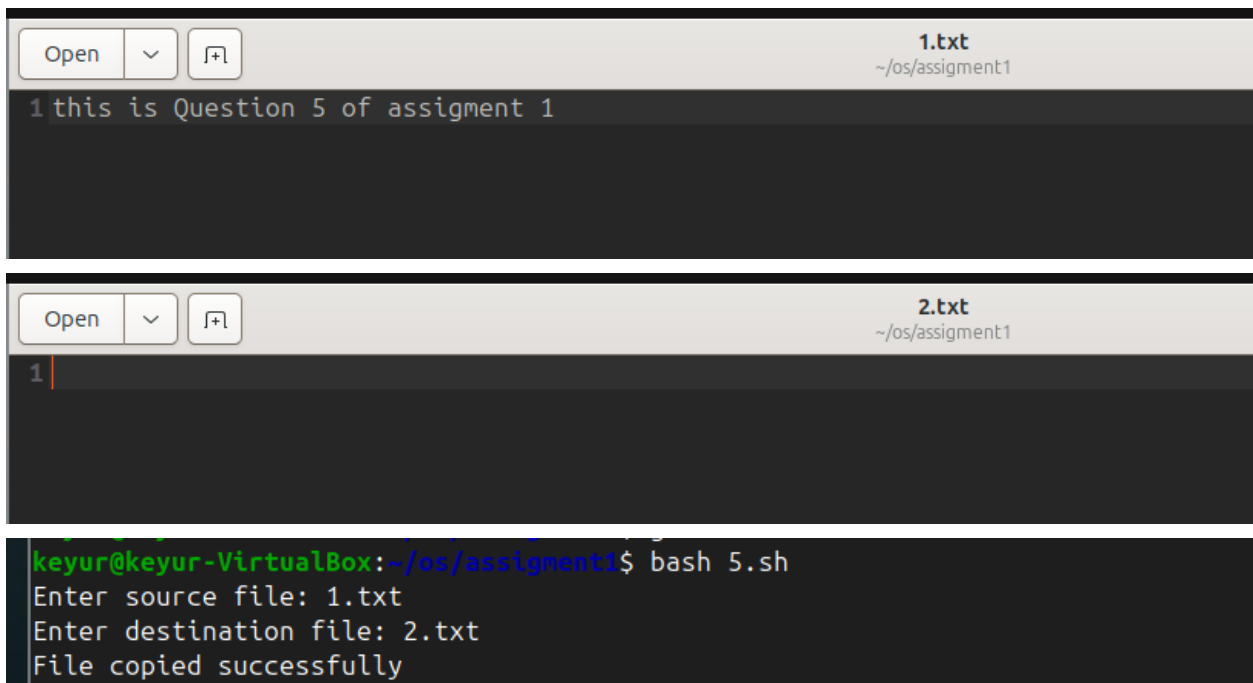
```
keyur@keyur-VirtualBox:~/os/assignment1$ gedit 4.sh
keyur@keyur-VirtualBox:~/os/assignment1$ bash 4.sh
Enter directory path: ../assignment1
Number of files: 8
keyur@keyur-VirtualBox:~/os/assignment1$ ls
1.sh 2.sh 3.sh 4.sh 5.sh 6.sh 7.sh 8.sh a1 a2 a3
keyur@keyur-VirtualBox:~/os/assignment1$
```

5. Shell Script Program to copy the contents of one file to another.

Code :

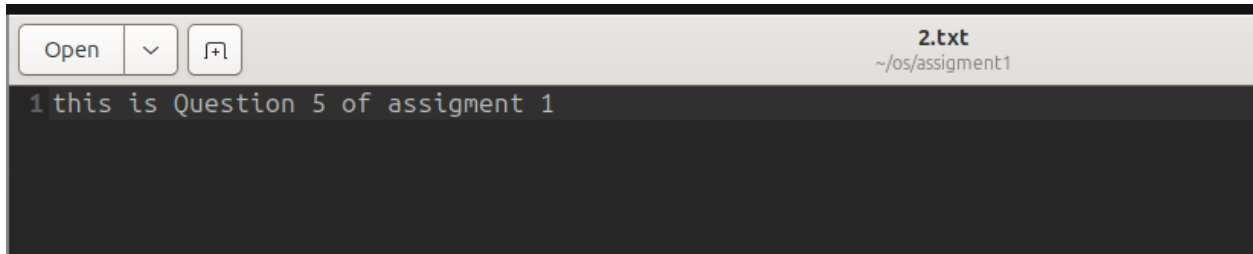
```
read -p "Enter source file: " src
read -p "Enter destination file: " dest

if [ -f "$src" ]; then
    cp "$src" "$dest"
    echo "File copied successfully"
else
    echo "Source file does not exist"
fi
```



The image shows a sequence of three screenshots from a virtual machine environment. The first screenshot shows a file editor window titled '1.txt' with the content '1this is Question 5 of assigment 1'. The second screenshot shows a file editor window titled '2.txt' which is currently empty. The third screenshot shows a terminal window where the user has executed 'bash 5.sh'. The script prompts for a source file (1.txt) and a destination file (2.txt), and then outputs 'File copied successfully'.

```
keyur@keyur-VirtualBox:~/os/assignment1$ bash 5.sh
Enter source file: 1.txt
Enter destination file: 2.txt
File copied successfully
```



6. Write a shell script to add two numbers supplied by the user and supplied as command-line Arguments

Code :

```
sum=$(( $1 + $2 ))
echo "Sum: $sum"
```

```
keyur@keyur-VirtualBox:~/os/assignment1$ bash 6.sh 1 2
Sum: 3
keyur@keyur-VirtualBox:~/os/assignment1$ bash 6.sh 5 9
Sum: 14
keyur@keyur-VirtualBox:~/os/assignment1$
```

7. Write a shell script to find out the biggest number from the given three numbers. Numbers are supplied by the command line argument.

Code :

```
if [ $1 -gt $2 ] && [ $1 -gt $3 ]; then
    echo "$1 is the biggest number"
elif [ $2 -gt $3 ]; then
    echo "$2 is the biggest number"
else
    echo "$3 is the biggest number"
fi
```

```
keyur@keyur-VirtualBox:~/os/assignment1$ bash 7.sh 5 19 3
19 is the biggest number
keyur@keyur-VirtualBox:~/os/assignment1$ bash 7.sh 20 8 0
20 is the biggest number
keyur@keyur-VirtualBox:~/os/assignment1$
```

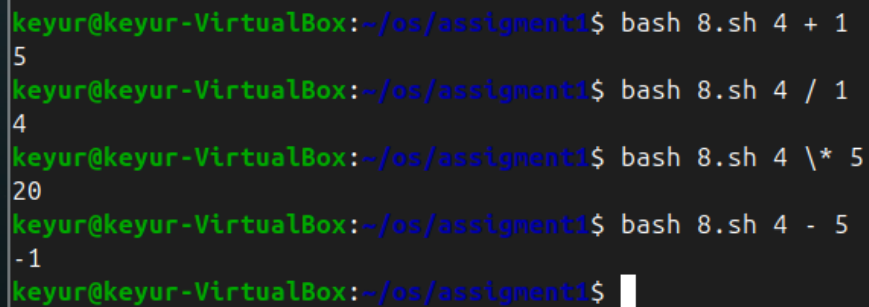
8. Implement a simple calculator. Numbers are supplied by the command line argument.

Code :

```
num1=$1
```

```
op=$2
num2=$3

case $op in
+) echo "$((num1 + num2))" ;;
-) echo "$((num1 - num2))" ;;
\*) echo "$((num1 * num2))" ;;
/)
    if [ $num2 -eq 0 ]; then
        echo "Division by zero error"
    else
        echo "$((num1 / num2))"
    fi ;;
*) echo "Invalid operator" ;;
esac
```



A terminal window with a dark background and green text. It shows the execution of a script named 8.sh with different operators. The prompt is keyur@keyur-VirtualBox:~/os/assignment1\$. The first command is bash 8.sh 4 + 1, which outputs 5. The second is bash 8.sh 4 / 1, which outputs 4. The third is bash 8.sh 4 * 5, which outputs 20. The fourth is bash 8.sh 4 - 5, which outputs -1. The final prompt is keyur@keyur-VirtualBox:~/os/assignment1\$ with a cursor.

```
keyur@keyur-VirtualBox:~/os/assignment1$ bash 8.sh 4 + 1
5
keyur@keyur-VirtualBox:~/os/assignment1$ bash 8.sh 4 / 1
4
keyur@keyur-VirtualBox:~/os/assignment1$ bash 8.sh 4 \* 5
20
keyur@keyur-VirtualBox:~/os/assignment1$ bash 8.sh 4 - 5
-1
keyur@keyur-VirtualBox:~/os/assignment1$
```